

# Erdős Problem 915 — Cheat Sheet (front)

one line per bound, page numbers into `main.pdf`  
conjectural

· **PROVED** exact value established  
**OPEN** general value open

**PARTIAL** partial or

## Simple graphs

$\ell_m(n)$  **edge**

**PROVED**

UP  $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ ,  $n \geq m \geq 2$  — Gomory–Hu tree, double-counted against Mader’s degree bound [p. 37–39]  
LO  $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$  — hub-and-spokes graph, spoke degree  $\leq m - 2$  [p. 40]

$k_m(n)$  **vertex**

**PARTIAL**

UP  $= \left\lfloor \frac{m(n-1)}{2} \right\rfloor$   $m \leq 4$  — edge bound does not transfer ( $\kappa \leq \lambda$  is the wrong direction); equal anyway up to  $m = 4$  (Leonard; Sørensen–Thomassen) [p. 6]  
LO  $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$  — same hub-and-spokes [p. 40]; Sørensen–Thomassen’s graph overtakes it from  $m = 5$ ,  $n \geq 14$  [p. 6]

$\ell_m^{\text{dir}}(n)$  **arc**

**OPEN**

UP  $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$   $m = 2$  — reachability skeleton, then the hub upper bound at  $m = 2$  [p. 47,49]  
LO  $M(n)$  — bidirected hub [p. 45] / one-way bipartite digraph [p. 46]

$k_m^{\text{dir}}(n)$  **vertex**

**OPEN**

UP  $M(n)$   $m = 2$  — sparse skeleton is already vertex-disjoint, plus deletion monotonicity [p. 47,49]  
LO  $\lfloor n^2/4 \rfloor$  — one-way bipartite digraph [p. 46]

## Multigraphs

$L_m(n)$  **edge**

**PROVED**

UP  $(m-1)(n-1)$ , every  $n, m$  — Gomory–Hu,  $n-1$  cuts each of capacity  $\leq m-1$  [p. 37]  
LO  $(m-1)(n-1)$  — any spanning tree at multiplicity  $m-1$  [p. 6]

$K_m(n)$  **vertex**

**PARTIAL**

UP  $2(n-1)$   $m = 3$  —  $\kappa(u, v)$  = copies of the edge + disjoint detours, capped together (incidence rank, then split) [p. 70,74–75]  
LO  $(m-1)(n-1)$  — doubled tree: no detours, so the whole cap goes to multiplicity [p. 75]

$L_m^{\text{dir}}(n)$  **arc**

**PROVED**

UP  $(m-1)M(n)$ , every  $n, m$  — peel a reachability skeleton off  $m-1$  times [p. 58]  
LO  $(m-1)M(n)$  — thickened bidirected tree / one-way bipartite digraph [p. 57]

$K_m^{\text{dir}}(n)$  **vertex**

**OPEN**

UP  $(m-1) \lfloor n^2/4 \rfloor + 4(m-1)^2(n-1)$  not tight [p. 83]  
LO  $n(n-1)(m+1-n)$  proposed,  $2 \leq n \leq m$  — thickened complete digraph, unique isomorphism class at  $n = 3, m = 6$  [p. 84]

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## Hypergraphs ( $r = 3$ )

$\ell_m^{(r)}(n)$  **edge**

**PARTIAL**

UP  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$  — cut induction on trace hypergraphs (Berge forbids the graph reduction) [p. 64]  
LO  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$   $m-1 \leq \binom{n-2}{r-2}$  — star hypertree / degree-capped hub [p. 65–66]

$k_m^{(r)}(n)$  **vertex**

**PARTIAL**

UP  $\left\lfloor \frac{n-1}{r-1} \right\rfloor$   $m=2$  — incidence graph is a forest ( $r|E| \leq n + |E| - 1$ ) [p. 67]  
LO  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$   $r \geq 3, m-1 \leq \binom{n-2}{r-2}$  — star hypertree [p. 69]

$\ell_m^{\text{dir}(r)}(n)$  **arc**

**OPEN**

UP  $\frac{(m-1)n(n-1)}{r-1}$  not tight — tail-head shadow digraph, first bounds [p. 85]  
LO  $n \cdot \min(m-1, \binom{n-1}{r-1})$   $n \geq r$  — bounded outgoing hyperedge count [p. 86]

$k_m^{\text{dir}(r)}(n)$  **vertex**

**OPEN**

UP  $\frac{(m-1)n(n-1)}{r-1}$  — outgoing cut never looks at interior vertices, same shadow argument [p. 85]  
LO  $n \cdot \min(m-1, \binom{n-1}{r-1})$   $n \geq r$  — same bounded-outdegree construction [p. 86]

## Multihypergraphs ( $r = 3$ )

$L_m^{(r)}(n)$  **edge**

**PARTIAL**

UP  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ , unchanged by repetition — same cut induction, repetition-blind [p. 64]  
LO  $\frac{(m-1)(n-1)}{r-1}$   $(r-1) \mid (n-1)$  — star hypertree, every hyperedge repeated  $m-1$  times [p. 69]

$K_m^{(r)}(n)$  **vertex**

**PARTIAL**

UP  $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$   $m=3$  — incidence-graph cycle rank via cut vertices / separating pairs [p. 70]  
LO  $\frac{(m-1)(n-1)}{r-1}$   $(r-1) \mid (n-1)$  — repeated star hypertree [p. 69]

$L_m^{\text{dir}(r)}(n)$  **arc**

**OPEN**

UP  $\frac{(m-1)n(n-1)}{r-1}$  — tail-head pair counting [p. 85]  
LO  $n(m-1)$   $n \geq r$  — every tail gets  $m-1$  copies; meets the bound exactly at  $n=r$  [p. 86]

$K_m^{\text{dir}(r)}(n)$  **vertex**

**OPEN**

UP  $\frac{(m-1)n(n-1)}{r-1}$  — carries over from the arc problem; general (any tail/head split) orientation shares the constant [p. 85,87,90]  
LO  $n(m-1)$   $n \geq r$  — same construction [p. 86]